Chapter 7 - Troubleshooting validation and maintenance

Practical source and control reference
29 October 2026

Use this guide to find a feature, understand its owner, make a bounded change, and check how that change is saved. Paths are relative to C:/Dropbox/Resume Website unless stated otherwise.

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Begin with evidence rather than resets

Record the route, browser, viewport, expected action, actual result and first relevant console/network error. Preserve the edited tab and export unsaved work. Reproduce in a separate review context. Avoid clearing storage, resetting the whole Git tree or changing several systems at once.

Search the active entry and its imports. A script named v2 or an old copy beside it is not proof of which file the current HTML loads. Main Music uses a dynamic import; the About fallback has an independent classic script so it can work if the 3D module fails.

A scene does not load

1. Check the first module error. A bare `three` specifier usually means the active import map did not resolve the dependency or a module loaded in the wrong context. Inspect the actual HTML served, not only the source you expected.
2. Check the requested GLB/data URL and response type. An HTML error page returned with status 200 is not a valid model or JavaScript file.
3. Confirm ignored external inputs are hydrated locally and inspect asset-host/CORS behavior separately from local paths.
4. Distinguish WebGL/driver failure from a network/module failure. Keep the native/text fallback usable in either case.
5. Reopen after a bounded fix and check errors, fallback navigation and input. Do not hide the error by permanently removing the fallback.

Sound is silent or out of sync

Check consent, site mute, zero volume, actual audio source and AudioContext/media readiness. A fetched MIDI file is not proof that a recorded soundtrack is playing. A loaded instrument is not proof that every scheduled note has an active source handle.

For paired media, inspect both clocks and the saved sync offset. Check beginning, middle, ending, pause/resume, seek and rate changes. Timing fixes should belong to the transport owner, not an unrelated UI loop that continually overwrites time or volume.

If a browser rejects audible play, preserve the site's real permission/user-gesture path. Do not turn a query parameter into permission. Device microphone and speaker/listening checks cannot be replaced by a successful fetch or muted headless run.

A setting seems ignored

1. Find the `data-setting` path and schema definition.
2. Check the current saved value and whether a legacy key wins normalization.
3. Verify the feature subscribes to that setting and has not captured a stale value once at startup.
4. Check whether Reduced motion overrides the effective animation while retaining its saved preference.
5. Use a field-specific reset or disposable context to compare defaults. Do not erase drawings/content to reset one preference.

A click acts on the wrong object

Check visual and hit bounds, nearest-hit occlusion, parent-space transforms and camera projection offsets. Ensure a pan/rotate/drag cannot leave a stale click candidate. Sticky-note editing, modals and private tool gestures should own input until they finish, including the trailing click after a drop.

Native keyboard controls should remain available for canvas-only actions where implemented. Page shortcuts must leave typing fields, IME composition, browser modifier chords and open dialogs alone.

Maintained commands and their effects

Use root `package.json` as the command inventory and `tools/VALIDATION.md` for prerequisites. Historical pass/failure counts in documents are not current results.

- `npm run review`: starts the loopback authoring/diagnostic preview.
- `npm run check:site-html`: checks generated regions/configuration without intentionally regenerating them.
- `npm run prepare:site-html`: writes generated header/import-map regions and worker config; coordinate all destinations first.
- `npm run lint`: source lint check.
- `npm test`: repository tests; use current required real fixtures and explicit scope rather than replacing them to obtain a pass.
- `npm run check:artifact` / `build:artifact`: inventory/build a new local package under the policy. Package creation is not release approval or deployment.
- `npm run preview:artifact`: static preview of a candidate. Local admin upload behavior is not available merely because static pages load.

Some check/build utilities write reports under output. Read a tool before assuming its name means it is read-only. Do not overwrite a retained release/evidence directory to rerun a check.

Review before publication

1. Confirm the exact candidate and the human's release authorization.
2. Include generated files, local source updates and required external inputs as one coherent version. A clean Git checkout alone may lack ignored media.
3. Check all affected direct routes and embedded projects, root/subpath URLs, real media decode, browser errors, fallback content and accessibility.
4. Check persistence/import/export, permission revocation, repeated navigation and resource cleanup.
5. Check asset licensing, fonts/vendor notices, hosting object keys and service-worker/cache behavior.
6. Keep a rollback and a known version of the previous coherent candidate. Do not publish an unrelated dirty working tree merely because one page looks correct.

Full mobile readiness remains a separate deferred gate. The mobile warning is not a substitute for touch/device testing. Automated/headless/source checks do not establish screen-reader, physical speaker, microphone or device GPU quality.

Keep this guide useful

Editable chapter sources are `docs/user-guide/`. The generator is `tools/build-user-guide.py`; PDFs are under `output/pdf/site-guide/`.

To regenerate one chapter with the bundled Python runtime:

```
python tools/build-user-guide.py docs/user-guide/03-about-cube-and-faces.md
```

Use the actual installed/bundled Python executable if the system python alias is unavailable. The script requires ReportLab and pypdf; visual review uses Poppler rendering. Check the generated pages before sharing the replacement.

When a control moves, update its UI path and source owner together. When an asset/format changes, document the source, export recipe, saved-ID compatibility and check routine. Keep historical evidence dated; link it instead of presenting it as fresh acceptance.

Remaining: browser-verify recipes, expand field-level indexes/screenshots, consolidate the detailed chapters if desired, and maintain them alongside future changes. The separate site program still includes architecture acceptance, settings/animation/keyboard work, controller extraction, content/asset cleanup and deferred mobile/model/release gates.